Homework (Chapter 8-9) **Due: Dec 6, 2022**

Student Name____________ Student No.______________ Points_________

1. Using the values for surface energy in Table 1 and Young's modulus in Table 2, rank in decreasing order the fracture toughness of single crystal MgO , Al_2O_3, SiC. Estimate a value for K_{IC} in each case, stating clearly any assumptions.

Table 1 Measured Surface Energies of Various Materials in Vacuum or Inert Atmospheres

	Temperature (℃)	Surface Energy (ergs/cm^2)
Liquids		
B_2O_3	900	80
FeO	1420	585
Al_2O_3	2080	700
0,13 Na_2O-0.13 CaO-0.74 SiO_2	1350	350
Solids		
Al_2O_3	1850	905
0.20 Na_2O-0.80 SiO_2	1350	380
NaCl (100)	25	300
$CaCO_3$ (1010)	25	230
MgO (100)	-196 (77K)	1500
Al_2O_3 ($101\bar{2}$)	25	6000
($10\bar{1}0$)	25	7300
(0001)	25	>40000
$MgAl_2O_4$ (100) (111)	25 25	3000 5000
SiC ($11\bar{2}0$)	25	20000

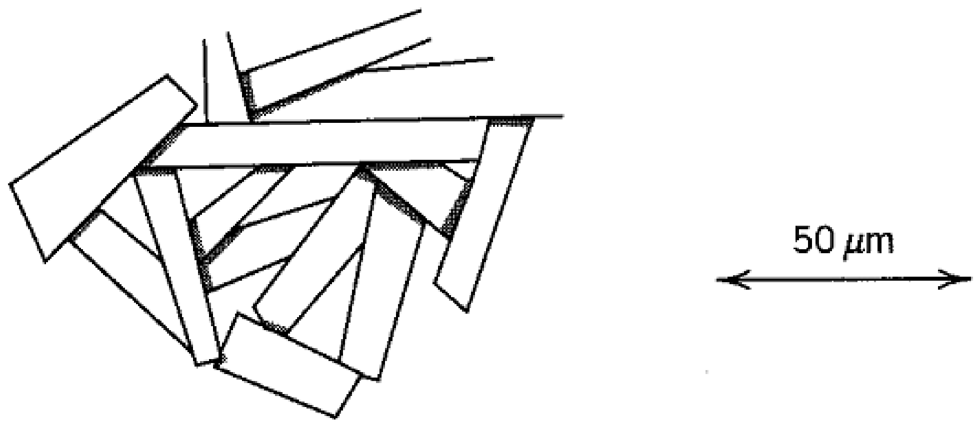
Table 2 Young's Modulus of Various Ceramics (at room temperature)

	Young's Modulus (GPa)
Fused silica	72
Soda-lime-silica glass	73
Pyrex borosilicate	69
Aluminosilicate glass	89
Single crystal Al_2O_3	380
Dense sintered Al_2O_3	366
Dense sintered Si_3N_4	304
MgO	207

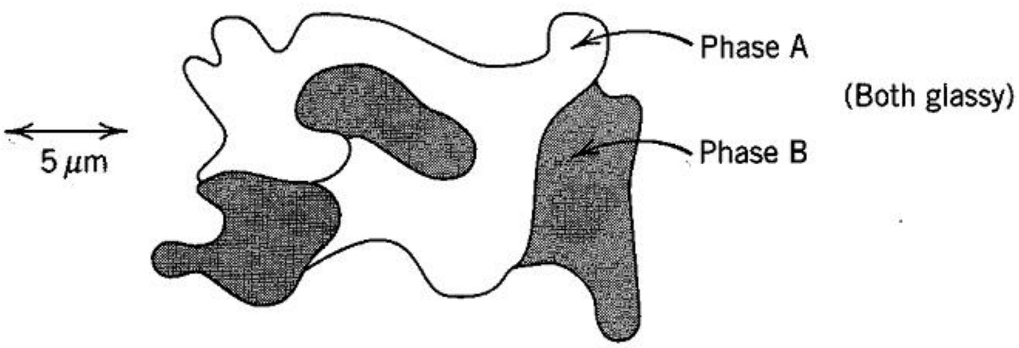
ZrO_2	138
SiC	414
TiC	430
B_4C	290
Al_3O_3N spinel	330
Diamond	1035

2. Which of the following polyphase ceramic microstructures might exhibit *R*-curve behavior, and which are likely to show conventional brittle fracture?.

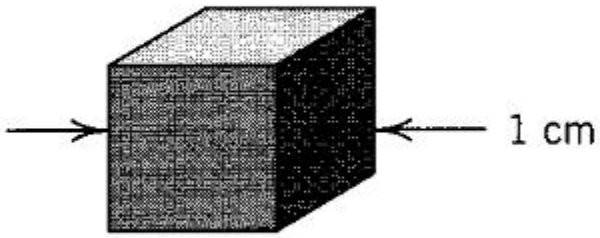
(a) Alumina with an elongated grain microstructure



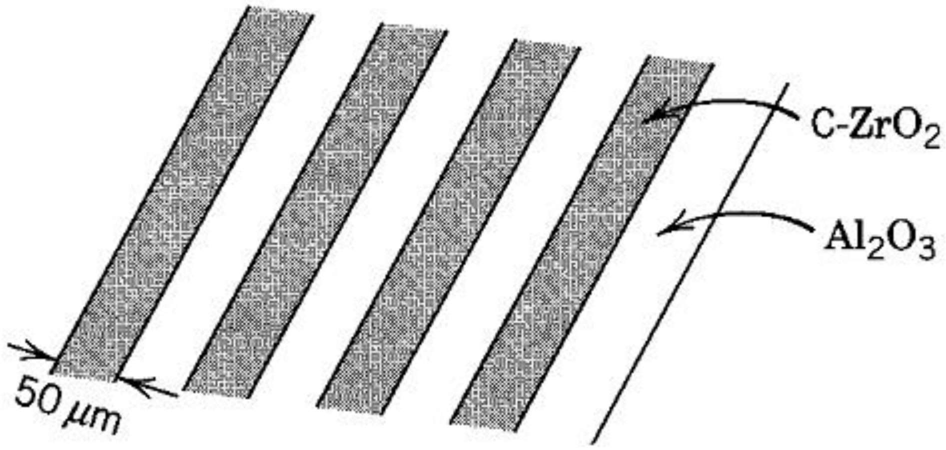
(b) Phase-separated glass



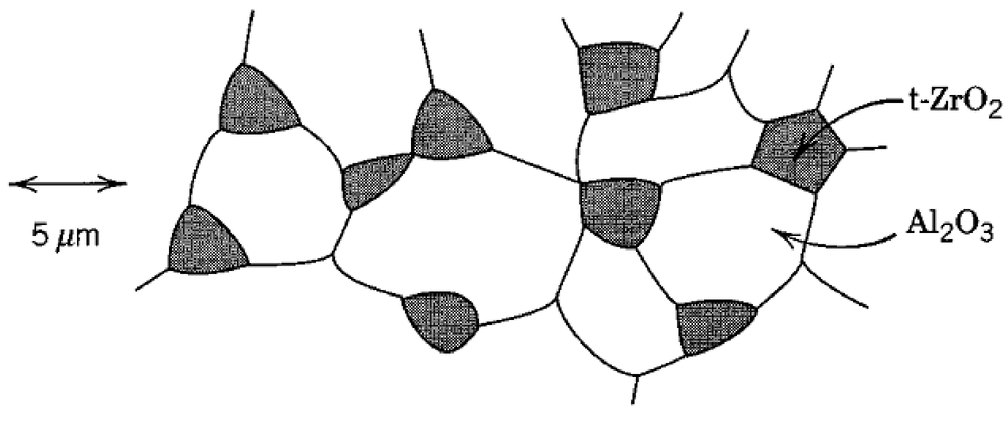
(c) Single crystal of MnZn ferrite



(d) Laminated alumina/cubic zirconia composite



(e) Particulate alumina-tetragonal zirconia composite



3. (a)The thermal conductivity of a single-crystal specimen is slightly greater than a polycrystalline one of the same material. Why is this so? (b) Briefly explain why the thermal conductivities are higher for crystalline than noncrystalline ceramics? For some ceramic materials, why does the thermal conductivity first decrease and then increase with rising temperature?
4. (a) Briefly explain why thermal stresses may be introduced into a structure by rapid heating or cooling. (b) For cooling, what is the nature of the surface stresses? (c) For heating, what is the nature of the surface stresses? (d) For a ceramic material, is thermal shock more likely to occur on rapid heating or cooling? Why?
5. In order to obtain satisfactory fit between a glaze and body, which is desirable for glaze after cooling to Room Temperature ?   Tension or Compression ? Why ? How to achieve ?